

Week 2 Assignment

Chapters 3-4

Name:

You may discuss homework with your classmates. But your answers here must be your own and you are required to show your work.

1. In labrador retrievers, a black coat (B) is dominant to chocolate coat (b) and Normal vision (N) is dominant over blindness (n). A heterozygous black lab that is blind is crossed with a homozygous black lab that is also blind.

a. What is the chance of this cross yielding a puppy of the genotype Bbnn?

$$nn \times nn = nn (100\%)$$

50% since the

homozygous cross ensures B

	B	b
B		✓
b		✓

 50%

"nn" and the hetero/homo cross has a 50% hybrid rate.

b. What is the chance of this cross yielding a puppy of the genotype bbnn?

0% since the homozygous BB parent has no "b" allele to offer.

	B	b
B		
B		

 (no bb)

c. What is the chance of this cross yielding a puppy of the genotype BbNn?

0% since neither parent has a "N" allele to offer.

	n	n
n		
n		

 (no N)

2. The term **epistasis** is used to refer to a situation in which the expression of a gene is influenced by another independently inherited gene. In labrador retrievers, the gene **B** determines how much pigment is made. The dominant allele **B** results in black fur and the recessive allele **b** results in brown (chocolate) fur.

A separate gene, **E**, codes for a protein that determines whether the pigment is deposited in the hair. (E = pigment deposited; e = pigment not deposited).

a. What genotypes are possible for a **black lab**?

BBEe (needs at least one B and E)
BBEE
BbEE (BxE)
BbEe

b. What genotypes are possible for a **chocolate lab**?

bbEE (needs at least one E)
bbEe

c. What genotypes are possible for a **yellow Lab**?

BBee (anything, but no "E" allele)
Bbee
bbee

d. In a cross between a black lab that is homozygous for both alleles and a yellow lab that is homozygous for both alleles, what would the genotype and phenotype of the offspring be?

phenotype = Black lab.

genotype = BBEE or BbEe

(we only know the parent is homozygous but not what the allele is.)

B
B
E
E

XXee = BBee
or
bbee.
BXEe

$$bbEE = 1/16$$

$$bbEe = 2/16$$

$$bb ee = 1/16$$

genotypes

$$BBEE = 1/16$$

$$BB Ee = 2/16 = 1/8$$

$$BB ee = 1/16$$

$$BbEE = 2/16 = 1/8$$

$$Bb Ee = 4/16 = 1/4$$

$$Bb ee = 2/16 = 1/8$$

e. Imagine you cross two heterozygous parents. What are the genotypic and phenotypic ratios of the offspring?

(dihybrid cross)

	BE	Be	bE	be
BE	BBEE	BbEe	bBEe	bbEe
Be	BBEe	BBee	bBEe	bb ee
bE	BbEE	BbEe	bbEE	bbEe
be	BbEe	Bbee	bbEe	bb ee

$$\text{yellow (ee)} = 4/16 \text{ or } 1/4$$

$$\text{Black (B x E x)} = 9/16$$

$$\text{Brown (bb E x)} = 3/16$$

phenotypes

3. What is meant by the phrase "genetic disease"? Is the common cold a genetic disease?

Explain.

A genetic disease is a disorder of health with a strong basis on a few genes, and high penetrance of the phenotype.

4. Use the following to respond to the three questions posed below:

- The **XX proband** exhibits the trait.
- Neither her **XY mate** nor her only sibling, and older **XY** individual, exhibits the trait.
- The proband has five children: The oldest is an **XY** individual, followed by an **XX** individual, then another **XY** individual and then identical twin **XX** individuals.
- Only the second oldest fails to exhibit the trait.
- Both parents of the proband show the trait.

I

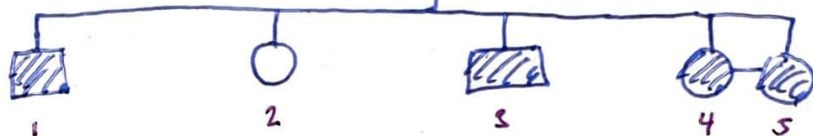


a. Construct a **pedigree** of the trait of the family (you can draw a picture and upload a picture or a scan or you can construct a pedigree using a digital method).

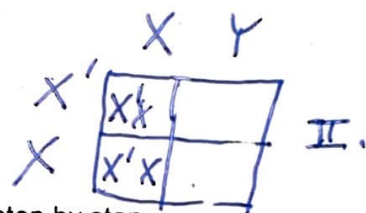
II



III



X'

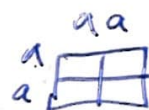


b. Determine how the trait is inherited (hint: for each pattern of inheritance, go step by step, through each generation to determine its likelihood).

1. X-linked - impossible since the XY of gen II is negative but recessive has a positive twin daughters, on III-4, 5

2. Autosomal dominant - possible.

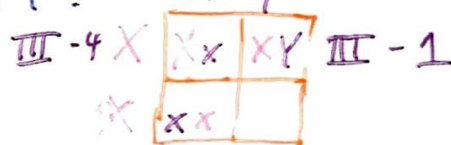
3. Autosomal recessive: impossible due to generation I. (all children would be affected)



4. X-linked dominant - possible. 5. Y-linked = no, since women have the trait too.

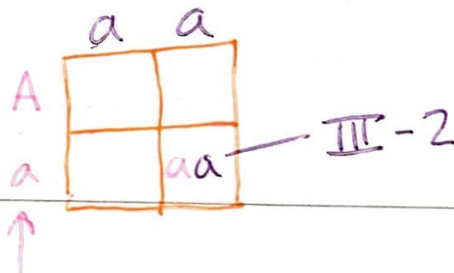
c. Can you deduce the **genotype of the proband's mate** for this trait? Explain why or why not.

1. if X-dominant: yes, the mate is XY: $X'Y$



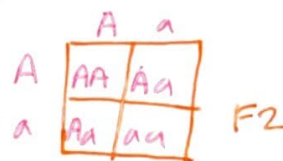
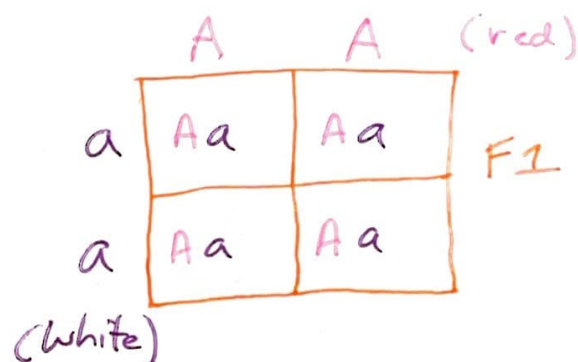
2. if autosomal dominant: yes, the mate is "aa,"

or homozygous recessive:



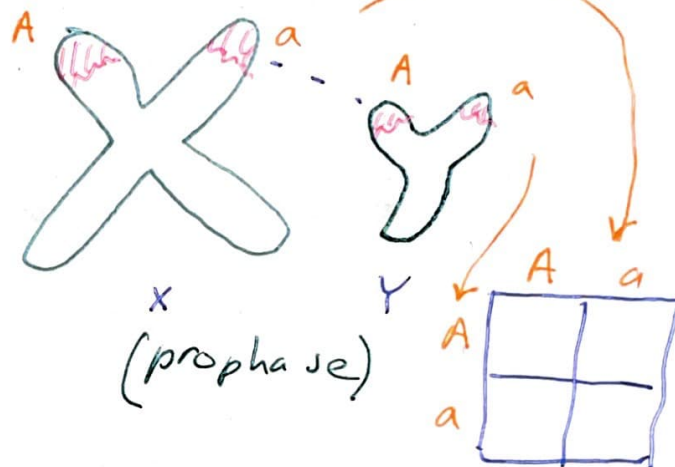
5. Imagine that a true-breeding **red-flowered snapdragon** is crossed to a true-breeding **white-flowered snapdragon**. All offspring are **pink-flowered**. Diagram this cross, including genotypes for all P2 and F1 phenotypes. What is the mode of inheritance?

This is incomplete dominance since the F1 phenotype has blended characteristics of both parents.



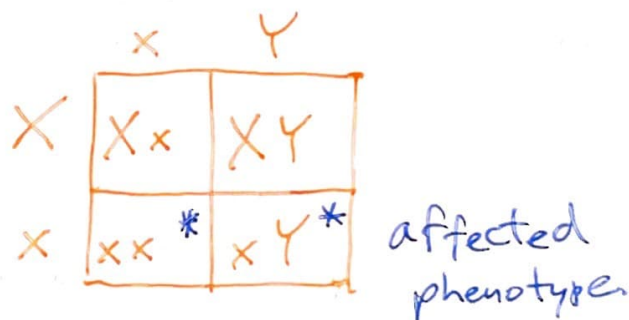
6. The X and Y chromosomes are structurally and genetically distinct. However, they do pair during meiosis at a small region near the tips of their short arms, indicating that the chromosomes are homologous in this region. If a gene lies in this region, will its pattern of transmission be more like that of a **sex-linked gene** or an **autosomal gene**? Explain your answer in 1-3 sentences.

This gene will be akin to an autosomal gene because it will have two alleles in each parent, rather than just one. This causes a greater range of possibilities during the Prophase stage.



7. What is the chance that a **color blind XY individual** and a **carrier XX individual** will produce:

- A color-blind child that is XY 25%
- A color-blind child that is XX 25%

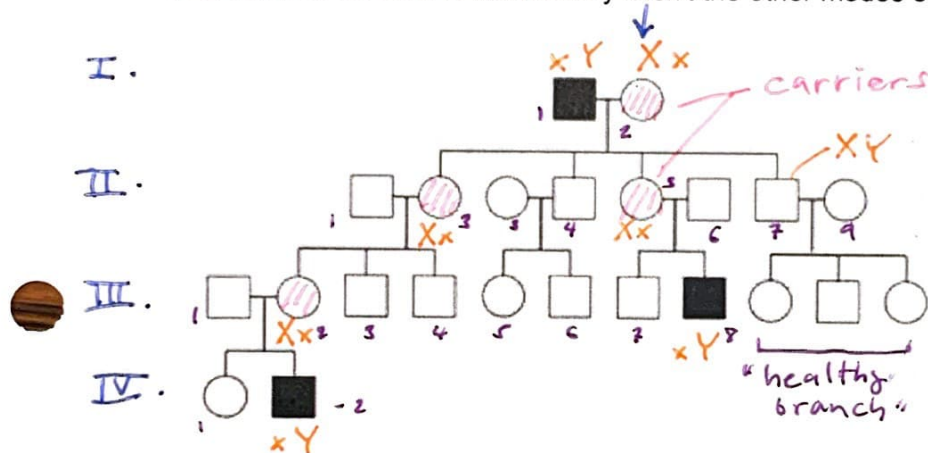


Since the CFTR protein is coded by two copies of the ABCC7 gene, it leads to the carriers being mostly unaffected.¹ (I'm one of them!). Specifically, even with one copy of functional ABCC7, there's a sufficient surplus of CFTR, albeit 50% normal. There may even be an evolutionary advantage of heterozygosity, since 4% of N. European populations are carriers, which is more than chance would predict.



1. Lehninger Biochemistry, 8th ed.

10. Analyze the following pedigree. Is the trait inherited as **autosomal dominant**, **autosomal recessive**, **X-linked dominant**, **X-linked recessive**, or **Y-linked**? How do you know? Explain your answer. Be sure to include why it isn't the other modes of inheritance.



This is X-linked recessive because 1) only males are affected, suggesting it's on the X chromosome, of which males only have one copy. This means there's no dominant "normal" X to balance it out. (see I-1, II-2, III-8, IV-2)

2) The trait disappears in generation II. This confirms it is recessive, since none of the new mates exhibit it, on II-1, II-3, or II-6.

3) It's not showing father $\rightarrow$ son transmission (Y-link)

4. It's not autosomal recessive since we'd expect equal male/female frequency.